Quantization-Aware Federated Learning with Site-Local Normalization for Cross-Dataset Brain Tumor MRI Classification on Edge Devices

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***Abstract*—Deploying automated brain tumor screening in low-resource clinical settings imposes three constraints at once: patient MRI cannot leave the institution that acquired it, inference must run on commodity hardware without a GPU or reliable connectivity, and the model must generalize to scanners it has never seen. These constraints are usually studied in isolation. We argue they interact. Federated averaging over sites with heterogeneous acquisition characteristics converges to a solution that is not a stationary point of any participant's local objective, and we hypothesize that such a solution is correspondingly less robust to the weight perturbation introduced by post-training integer quantization—so that privacy-preserving training silently costs deployability. We propose QAF-Net, which addresses the interaction directly by augmenting the local federated objective with a per-output-channel INT8 fake-quantization regularizer, so that aggregation occurs between weight sets that already lie close to an integer-representable grid. We further keep batch-normalization parameters site-local, on the grounds that their running statistics encode precisely the scanner intensity profile that constitutes the domain shift, and re-estimate them at a new site from unlabeled local scans. A duplicate audit shapes the evaluation: 55.2% of the images in the most widely used aggregated brain-tumor benchmark are near-duplicates of its constituent public sources, and after deduplication only two public cohorts, from China and Bangladesh, retain substantial subtype labels. We therefore federate four real cohorts with heterogeneous label spaces, admitting cohorts without subtype labels through a superset loss, and evaluate leave-one-site-out. The hypothesis is not supported: a centrally trained model loses 8.6 points to INT8 quantization on the Bangladeshi cohort, more than any federated method does on average, and at the strength tested the quantization regularizer leaves the distance of the weights from the integer grid unchanged. Normalization is what matters. Re-estimating batch-normalization statistics from unlabeled target scans raises the held-out INT8 accuracy of federated averaging on the Chinese cohort from 76.7% to 93.3% and shrinks its spread across seeds from 17.7 to 2.0 points; with it, QAF-Net reaches 94.7% in a 606 KB model that loses 0.15 points to quantization, although that advantage over FedProx cannot be credited to the regularizer. One failure is shared by every federated method: a class held by a single client is lost, and the best of them recognizes 3% of the normal scans in the held-out mixed cohort.**

***Index Terms*—federated learning, TinyML, quantization, brain tumor classification, magnetic resonance imaging, domain shift, edge computing, digital image processing**

I. Introduction

Gliomas, meningiomas and pituitary adenomas account for a large share of primary intracranial tumors, and their differentiation from magnetic resonance imaging (MRI) is a routine but expertise-dependent task. Convolutional neural networks now classify these three types with reported accuracies above 98% on public benchmarks [1]–[3], and a substantial survey literature documents the maturity of the approach [4]. Despite this, deployed clinical decision support for brain tumor triage remains rare, particularly in the low- and middle-income settings where the shortage of neuroradiologists is most acute.

The gap is not one of classification accuracy. It is that the conditions under which those accuracies are obtained do not hold in deployment. We identify three constraints that a fielded system faces simultaneously.

Data cannot be centralized. Patient imaging is subject to institutional and regulatory restrictions on transfer. The standard practice of pooling several public datasets into a single training corpus is unavailable to a hospital consortium that wishes to train on its own records. Federated learning [5] was introduced for exactly this situation and has been applied to medical imaging [6]–[8].

Inference must run on constrained hardware. A district hospital may have neither a GPU nor dependable connectivity to a remote inference service. This is the domain of TinyML [9], [10], where models are compressed to integer arithmetic and deployed on microcontroller-class devices, typically via post-training quantization [11], [12].

The model must survive an unseen scanner. Public brain tumor datasets aggregate images across manufacturers, field strengths and acquisition protocols. A model that is validated on a random split of such a corpus is validated under the assumption that deployment images are drawn from the same mixture, which for a new hospital they are not.

These three requirements are ordinarily treated as separable engineering concerns to be composed in sequence: train federated, then quantize, then hope the result transfers. **Our central claim is that this composition is not benign.**

A. The interaction

Federated averaging returns a parameter vector $\bar{w} = \sum_k p_k w_k$ formed from client solutions w_k that, under non-identically distributed data, minimize materially different local objectives. The average is in general not a stationary point of any client's loss, nor of their mixture. Post-training quantization perturbs the deployed weights by δw, and to second order the induced loss increase is $\frac{1}{2}\,\delta w^\top H\,\delta w$ for local curvature H. A solution that does not sit in a well-conditioned local minimum has no reason to absorb that perturbation as gracefully as a centrally trained one does. We therefore expect the FP32→INT8 degradation to be *larger* after federated training than after centralized training, with the excess growing in the heterogeneity of the federation.

This interaction is invisible to both parent literatures. Federated learning research reports full-precision accuracy; TinyML research quantizes centrally trained models. We are not aware of a study that measures the deployment-time quantization cost *attributable to federation* in medical imaging. We treat it as a hypothesis to be tested rather than an established effect, and we report the outcome either way.

B. Contributions

1) **A duplicate audit of the standard benchmark.** The most widely used aggregated brain tumor MRI dataset is a merge of three public sources, one of which proves to be largely a redistribution of another. We show that a perceptual-hash detector—the obvious tool—is unusable on brain MRI because every slice shares the same global structure, and instead calibrate a normalized cross-correlation criterion against an independent-cohort negative control. In the current releases, 71.8% of SARTAJ consists of copies of CHENG, 45.5% of BR35H has a near-duplicate within the cohort, and 55.2% of the aggregated benchmark duplicates its sources (Sections III, VI).
2) **QAF-Net**, a federated objective containing a per-output-channel INT8 fake-quantization regularizer that constrains each client's update toward an integer-representable grid during training, so that the aggregated model is quantization-robust by construction rather than by post-hoc repair (Section IV-B). We also give a direct test of whether the regularizer acts, and find that at $\lambda = 0.01$ it does not (Section VI-C).
3) **Site-local normalization with deployable adapters.** Batch-normalization statistics are excluded from aggregation and re-estimated at a new site from unlabeled scans using forward passes alone—no labels, no gradients, and no data leaving the institution. This is the largest effect we measure (Sections IV-C, VI-B).
4) **A label-space heterogeneous federation.** Real sites do not agree on a label space: a binary tumor-detection cohort cannot supply subtype labels, and two subtype-labelled cohorts contain no normal scans. After deduplication this is not an edge case but the only way to federate more than two public cohorts. We admit such clients through a superset loss that reduces to cross-entropy when the label is a singleton (Section IV-D).
5) **A leave-one-site-out evaluation with deployment metrics**, reporting integer model size, multiply–accumulate count, peak tensor-arena memory and CPU latency alongside accuracy and subtype accuracy, over four public cohorts (Section V), and a test of the interaction hypothesis, which our data do not support (Section VI-C).

II. Related Work

A. Deep learning for brain tumor MRI classification

Deepak and Ameer [1] applied GoogLeNet with transfer learning to the three-class figshare cohort, reporting 98% mean accuracy under patient-level five-fold cross-validation. Sultan et al. [2] studied block-wise fine-tuning of pretrained CNNs for the same task, and Zulfiqar et al. [3] evaluated the EfficientNet family, with EfficientNetB2 reaching 98.86% test accuracy and Grad-CAM [13] used for visual explanation. The survey of Amin et al. [4] catalogues the field and identifies dataset variation and generalization as recurring open problems.

Two properties of this literature matter here. First, evaluation is almost universally on a random or patient-level split of a *single* corpus, so the reported figure estimates in-distribution performance rather than transfer to a new institution. Second, the architectures evaluated are large—GoogLeNet, ResNet-50, the EfficientNet family—and their deployment cost is not reported, because deployment is not the object of study.

B. Federated learning in medical imaging

McMahan et al. [5] introduced federated averaging. Its degradation under non-identically distributed clients motivated FedProx [14], which adds a proximal term penalizing divergence from the global model, and FedBN [15], which keeps batch-normalization layers local to each client and is explicitly targeted at feature-distribution skew. In medical imaging specifically, Sheller et al. [6] demonstrated multi-institutional collaboration without data sharing, Li et al. [8] addressed privacy-preserving brain tumor segmentation, and Rieke et al. [7] survey the clinical prospects. This body of work reports full-precision performance; the deployed artifact is assumed to be the trained network itself.

Our site-local normalization follows FedBN's insight, and we adopt it as a baseline. We extend it in the deployment direction: because the site-local parameters are small, they can be shipped as adapters, and because batch-normalization statistics can be estimated without labels [16], a new site can calibrate them from unlabeled scans.

TABLE I
MRI COHORTS BEFORE AND AFTER DEDUPLICATION (SECTION III-B).

Site	Provenance	Loaded	Kept	Classes
CHENG	China, 2 hospitals [20]	3,064	3,064	gli/men/pit
PMRAM	Bangladesh [21]	6,056	6,012	gli/men/pit
SARTAJ	Mixed public [22]	3,264	544	gli/men/pit/none
BR35H	Independent, binary [23]	3,000	1,908	tumor/none
Audit reference only, never a client:				
MERGED	Merge of the above [24]	7,200	—	gli/men/pit/none

C. Quantization, TinyML, and quantization in federated settings

Post-training and quantization-aware training for integer-only inference were formalized by Jacob et al. [11] and surveyed by Krishnamoorthi [12]; MobileNetV2's inverted residual design [17] remains a standard efficient backbone. TensorFlow Lite Micro [9] and the MLPerf Tiny benchmark [10] established the microcontroller deployment target, where flash and SRAM budgets, not FLOPs, are typically binding.

Quantization has been studied within federated learning, but with a different objective. FedPAQ [18] and related methods quantize the client *updates in transit* to reduce communication cost; the model itself remains full-precision, and the quantizer is discarded after aggregation. Our concern is orthogonal and complementary: we quantize the *deployed model*, and we ask what federation does to its quantizability. To our knowledge the combination—a federated objective explicitly shaped so that the resulting global model survives integer quantization for edge deployment—has not been examined for medical image classification.

D. Learning from partial labels

Where a client supplies a set-valued rather than exact label, the appropriate objective is the superset loss of Cour et al. [19]. We use it to admit a binary tumor-detection cohort into a subtype-classification federation without discarding its positive cases.

III. MATERIALS

A. Cohorts

We use four public MRI cohorts, summarized in Table I, chosen so that federated clients correspond to *genuinely distinct acquisitions* rather than to an artificial partition of a single corpus. Simulating non-identical distributions by Dirichlet-splitting one dataset, as is common, reproduces label skew but not the feature skew—scanner, field strength, protocol, population—that a real deployment encounters.

CHENG [20] comprises 3,064 T1-weighted contrast-enhanced slices from 233 patients and is the only cohort distributing patient identifiers. All CHENG splits are therefore grouped by patient: adjacent slices of one lesion are highly correlated, and permitting them to straddle a train/test boundary is a well-known source of optimistic bias. The remaining cohorts distribute image files without patient identifiers, which we note as a limitation. PMRAM [21] was acquired across hospitals in Bangladesh with radiologist involvement and is, to our knowledge, the only large public brain tumor cohort acquired independently of the figshare lineage; our audit confirms that it shares no images with any other source (Section III-B).

B. Cross-source duplication

The widely used aggregated benchmark [24] is a merge of CHENG, SARTAJ and BR35H, and SARTAJ is itself partly re-derived from CHENG. Treating these as distinct sites without removing the overlap would allow a model to be tested on images it was trained on.

Perceptual hashing is the conventional detector and is *inapplicable here*. Every brain MRI slice presents the same global structure—a bright ovoid on a dark field—so a 64-bit difference hash places independent patients within a few bits of one another. On synthetic data with known ground truth, a difference hash at Hamming distance ≤ 5 flagged approximately 78% of genuinely independent images as duplicates.

We instead compute normalized cross-correlation on mean-subtracted 64×64 thumbnails, $\rho_{ij} = \tilde{v}_i^\top \tilde{v}_j$ with $\tilde{v} = (v - \bar{v})/\|v - \bar{v}\|_2$, evaluated as a single matrix product, together with an exact MD5 check over stored pixels. The decision threshold τ is calibrated rather than assumed: CHENG (China) and PMRAM (Bangladesh) share no provenance, so their cross-correlation distribution is a negative control for "different patient, different scanner," and any admissible τ must exceed its upper tail. We report the resulting threshold, the full overlap matrix (Table II) and a sensitivity analysis over τ below, and we deduplicate in provenance order: an image is removed from a cohort if it matches an image already retained in a higher-priority cohort. Within every cohort we additionally keep only the first image of each near-duplicate group, and discard both members of any within-cohort pair whose labels disagree, since the correct label cannot be determined. We report how many images each rule removes, and how often a cross-cohort copy disagrees with the label of its source—a direct measure of label noise in the derived datasets.

Applied to the current releases, the audit is decisive (Table II). The calibrated threshold is $\tau = 0.999$, and the flagged fractions barely move with it: for SARTAJ, measured against the other cohorts, they range from 83.9% at $\tau = 0.99$ to 83.5% at $\tau = 0.9995$. Of SARTAJ's 3,264 images, deduplication removes 2,291 as copies of CHENG, 394 as exact copies within SARTAJ, 31 as within-cohort near-copies and 4 as label conflicts, leaving 544, of which only 235 carry a tumor subtype. Only 14 of the 2,345 cross-cohort copies (0.6%) disagree with the label of their CHENG source. BR35H is internally redundant: 45.5% of its images have a near-duplicate inside the cohort, and 727 are removed as within-cohort copies. Of the aggregated benchmark's 7,200 images, 55.2% duplicate CHENG, SARTAJ or BR35H, and 23.0% have a near-duplicate within the benchmark itself. PMRAM shares no images with any other source.

C. Label space and preprocessing

Labels are unified to $\{\text{glioma}, \text{meningioma}, \text{pituitary}, \text{notumor}\}$. The deduplicated cohorts observe different subsets of this space: CHENG and PMRAM contain the three tumor subtypes but no normal scans, SARTAJ contains all four classes, and BR35H contains normal scans and tumors of unknown subtype. With SARTAJ reduced to 235 subtype-labelled images, no federation of three or more clients with a shared label space remains. The primary experiments therefore use all four classes with all four cohorts as clients, and BR35H's tumor images enter through the superset loss of Section IV-D.

The cohorts also differ in acquisition. CHENG is contrast-enhanced T1-weighted, and the PMRAM slices we inspected are consistent with T1-weighted acquisition, whereas BR35H and part of SARTAJ include T2-weighted and FLAIR slices. Every normal training image therefore comes from cohorts that include those sequences. In the release we use, PMRAM slices are also stored rotated by $\pm 90^\circ$ relative to the other cohorts.

Slices are min–max normalized per image to `uint8`, resized to 144×144, and stored once. Training draws random 128×128 crops with random quarter-turn rotations, horizontal flips, and mild brightness and contrast jitter, so that in-plane orientation cannot act as a site signature; evaluation uses the center crop. Inputs are scaled to $[-1, 1]$ and replicated to three channels for ImageNet-pretrained backbones.

IV. METHODOLOGY

A. Problem setting

K institutions hold private datasets $\mathcal{D}_k = \{(x_i, S_i)\}_{i=1}^{n_k}$, where $S_i \subseteq \{1, \dots, C\}$ is a set-valued label, a singleton in the fully labeled case. Writing $n = \sum_k n_k$ and $p_k = n_k/n$, the federation seeks

$$\min_{\theta} \sum_{k=1}^{K} p_k \, \mathcal{L}_k(\theta), \tag{1}$$

subject to two deployment requirements absent from the standard formulation: the parameters θ must remain accurate after per-channel INT8 quantization, and evaluation is on a site $k^\star \notin \{1, \dots, K\}$ that contributed no data.

We partition parameters as $\theta = (\phi, \psi_1, \dots, \psi_K)$, where ϕ collects all weights that are aggregated and ψ_k the site-local normalization parameters of client k (Section IV-C).

B. Quantization-aware federated objective

Let Θ_q index the convolutional and fully connected kernels. For kernel W_j with output channels indexed by o, symmetric per-channel INT8 quantization uses scale

$$s_{j,o} = \frac{\max_{\cdot} |W_j[\cdot, o]|}{2^{b-1} - 1}, \qquad b = 8, \tag{2}$$

and the round-trip operator

$$Q_j(W_j)[\cdot, o] = s_{j,o} \cdot \mathrm{round}(W_j[\cdot, o]/s_{j,o}) \,. \tag{3}$$

Depthwise kernels are quantized along their input-channel axis, matching the deployment runtime's convention. We define the quantization residual

$$\mathcal{R}_q(\theta) = \sum_{j \in \Theta_q} \frac{1}{|W_j|} \, \| W_j - \mathrm{sg}[Q_j(W_j)] \|_2^2 \,, \tag{4}$$

where $\mathrm{sg}[\cdot]$ denotes stopping the gradient. Treating the quantized value as a constant target makes the gradient of (4) point from each weight toward its nearest representable grid point, which is the behavior we want: the term shapes the weights without attempting to differentiate through rounding.

The local objective at client k is then

$$\mathcal{L}_k(\theta) = \underbrace{\frac{1}{n_k} \sum_{(x,S) \in \mathcal{D}_k} \ell(S, f(x; \theta))}_{\text{superset cross-entropy}} + \frac{\mu}{2} \|\phi - \phi^t\|_2^2 + \lambda \, \mathcal{R}_q(\theta), \tag{5}$$

with ϕ^t the global parameters broadcast at round t. The second term is the FedProx proximal penalty [14], controlling client drift. The third is our contribution.

The intent of $\lambda \mathcal{R}_q$ is to make every client's solution lie close to the INT8 grid, so that aggregation combines weight sets that are individually quantization-friendly and the average inherits that property. This is distinct from quantization-aware training in the centralized sense, where the forward pass is quantized: here the forward pass remains full-precision and only the parameter geometry is constrained, which keeps the federated optimization well behaved and adds one elementwise operation per kernel per step. Setting $\lambda = 0$ recovers FedProx exactly, so the method contains its own ablation.

C. Site-local normalization

Batch-normalization [25] maintains running estimates of per-channel activation mean and variance. In multi-site MRI these statistics track the intensity profile of the acquiring scanner—exactly the nuisance factor constituting the domain shift. Averaging them across sites produces statistics describing no scanner in particular.

We therefore exclude the batch-normalization affine parameters and running statistics from aggregation, following FedBN [15]: each client retains its own ψ_k across rounds, and only ϕ is averaged,

$$\phi^{t+1} = \sum_{k=1}^{K} p_k \, \phi_k^{t+1}, \qquad \psi_k^{t+1} \text{ retained locally.} \tag{6}$$

Deployment at an unseen site $k^\star$ requires normalization parameters that no client can supply. We evaluate two mechanisms. The first sets $\psi_{k^\star} = \sum_k p_k \psi_k$, the size-weighted client average. The second *recalibrates*: the model is run in training mode over unlabeled scans from the target site, updating only the running statistics. No labels are required, no gradient is computed, and no image leaves the site. For our backbone ψ covers 7,040 channels, or 28,160 scalars (110 KB at FP32, 6.8% of the parameters). Integer conversion folds these parameters into the preceding convolutions, so adapting

a deployed model to a new site means re-estimating them and re-running the conversion, a CPU-side step that needs no labels.

D. Label-space heterogeneous clients

A binary tumor-detection cohort knows that a positive scan contains a tumor but not which subtype. Discarding those images wastes the majority of the cohort. Following Cour et al. [19] we use the superset loss

$$\ell(S, z) = -\log \sum_{c \in S} \frac{\exp(z_c)}{\sum_{c'} \exp(z_{c'})} = -\log \sum_{c \in S} \mathrm{softmax}(z)_c, \tag{7}$$

which assigns a positive BR35H scan the label $S = \{\mathsf{glioma}, \mathsf{meningioma}, \mathsf{pituitary}\}$ and reduces to standard cross-entropy whenever $|S| = 1$. The softmax remains over the full label space, so the global model stays calibrated across all C classes even though no single client observes all of them. This is exactly the regime the deduplicated cohorts impose: CHENG and PMRAM never observe notumor, and BR35H never observes a subtype.

E. The QAF-Net algorithm

QAF-Net

input: rounds T, local epochs E, rates η, μ, λ
initialize ϕ^0 from ImageNet; $\psi_k^0 \leftarrow \psi^0$ $\forall k$
for $t = 0$ to $T - 1$ **do**
  for each client k **in parallel do**
    $\theta \leftarrow (\phi^t, \psi_k^t)$ *// site keeps its own BN*
    reset local optimizer state
    for E epochs **do**
      $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_k(\theta)$ using (5)
    end for
    $(\phi_k^{t+1}, \psi_k^{t+1}) \leftarrow \theta$
  end for
  $\phi^{t+1} \leftarrow \sum_k p_k \phi_k^{t+1}$
end for
$\psi^\star \leftarrow \sum_k p_k \psi_k^T$ *// or recalibrate on target*
return INT8 quantization of $(\phi^T, \psi^\star)$ via (3)

Algorithm IV-E states the procedure. Local optimizer state is reset each round so that clients remain stateless, as the federated setting requires. Communication per round is $|\phi|$ parameters, identical to federated averaging: the quantization regularizer is computed locally and adds no message.

V. EXPERIMENTAL SETUP

A. Protocol

Evaluation is **leave-one-site-out**: the federation trains on the remaining cohorts and is tested on the entirely held-out one. This measures what a new hospital would experience on day one. The two primary folds hold out CHENG and PMRAM, tested on their full sets of 3,064 and 6,012 tumor slices, and train the three remaining cohorts as heterogeneous clients. A secondary fold holds out SARTAJ: at 544 images it is small, but it is the only test cohort that contains normal scans (309 of them). BR35H, which has no subtype labels, is always a training client. Each fold also keeps a validation split of the training sites, grouped by patient for CHENG; accuracy on it is the internal-validation figure that published work usually reports.

B. Architecture

We use **MobileNetV2** [17] with width multiplier 0.35 at 128×128 input, initialized from ImageNet, followed by global average pooling, dropout of 0.3 and a four-way linear classifier: 415,332 parameters and 18.9 M multiply–accumulates per image. A smaller network trained from scratch for the sub-256 KB regime is left to future work.

C. Baselines

Centralized pools all training sites; it is conventionally the accuracy upper bound, though it is precisely what the privacy constraint forbids. *Local-only* trains at a single site—what an institution obtains by declining to collaborate. Both train for 25 epochs, are run for one seed, and keep the batch-normalization statistics of their own training data. Between them we compare *FedAvg* [5], *FedProx* [14] ($\mu = 0.01$), *FedBN* [15], and *QAF-Net* ($\mu = 0.01$, $\lambda = 0.01$). Every method is evaluated at FP32 and after full-integer INT8 conversion, and the primary comparison is the *difference* between the two, which isolates quantization robustness from raw accuracy.

D. Training configuration

Adam at 10^{-3}, batch size 32, $T = 25$ rounds of $E = 2$ local epochs. Each trained federated model is evaluated with both mechanisms of Section IV-C. Recalibration runs up to 100 unlabeled forward passes (batch 32, momentum 0.9) over the held-out cohort, 17 for SARTAJ. It is therefore transductive: it sees the images it is evaluated on, though never their labels; a deployment would use a separate unlabeled sample from the site. Post-training quantization uses 100 representative batches drawn from the pooled training sites, with int8 input and output tensors. Every federated configuration is run for three seeds on every fold; we report mean and standard deviation. Training runs on Kaggle NVIDIA T4 GPUs; latency is measured on the CPU, since a GPU figure is not a meaningful proxy for edge deployment.

E. Metrics

Accuracy, macro-F1 over the classes present in the test set, confusion matrices, and one-versus-rest AUC. Because the CHENG and PMRAM folds contain only tumors, we also report *subtype accuracy*, in which tumor images are classified by the three tumor logits alone, and the fraction of tumors assigned to notumor. Subtype accuracy is largely insensitive to the sequence confound described in Section III, which acts mainly through the notumor output. Deployment is characterized by `.tflite` artifact size, multiply–accumulate count, estimated peak tensor arena, and mean and 95th-percentile single-image CPU latency. Grad-CAM [13] attribution maps are inspected on held-out-site images to check whether predictions are driven by lesion regions rather than by acquisition artifacts.

TABLE II

DUPLICATION AT $\tau = 0.999$: PERCENTAGE OF THE ROW COHORT'S IMAGES THAT HAVE A NEAR-DUPLICATE IN THE COLUMN COHORT. THE DIAGONAL COUNTS NEAR-DUPLICATES WITHIN THE COHORT. OF THE AGGREGATED BENCHMARK, 55.2% OF IMAGES DUPLICATE AT LEAST ONE SOURCE AND 23.0% HAVE A NEAR-DUPLICATE WITHIN THE BENCHMARK.

	CHENG	PMRAM	SARTAJ	BR35H
CHENG	0.0	0.0	74.8	0.1
PMRAM	0.0	1.3	0.0	0.0
SARTAJ	71.8	0.0	22.2	11.8
BR35H	0.1	0.0	12.3	45.5
MERGED	24.9	0.0	38.6	23.9

VI. RESULTS

A. Duplicate audit

Table II gives the overlap matrix at the calibrated threshold and Section III-B the counts behind it. Two numbers set the scope of everything that follows: SARTAJ shrinks from 3,264 images to 544, and only CHENG and PMRAM keep more than a few hundred subtype-labelled images. A quarter of the aggregated benchmark duplicates CHENG, so a model trained on it and tested on CHENG would be tested partly on its own training images.

B. Held-out accuracy and normalization

Table III and Fig. 1 give the leave-one-site-out results. How the held-out site obtains its batch-normalization statistics matters more than which federated algorithm trained the model.

With averaged client statistics, the federated models are unstable across seeds. On the CHENG fold FedAvg reaches 56.4%, 88.5% and 85.1% INT8 accuracy on its three seeds and FedProx 40.5%, 94.2% and 87.2%; standard deviations reach 29 points. Recalibrating the statistics on unlabeled CHENG scans brings every method to 93.3–94.7% with a standard deviation of at most 2.0 points, a gain of 16.6 to 20.7 points. On PMRAM, recalibration lifts FedAvg by 19.8 points to 77.3% and cuts the spread of FedBN and QAF-Net from about 20 points to 2–3, but it is not uniformly beneficial: FedProx falls from 84.9% to 64.9%, and after recalibration it sends 53% of meningiomas to glioma and 27% of pituitary tumors to notumor (Fig. 3). With recalibrated statistics QAF-Net is the most accurate method on CHENG (94.7%) and FedBN on PMRAM (78.3%), but on CHENG the four federated methods are within about one standard deviation of one another. The FP32 one-versus-rest AUC of the recalibrated models stays between 97.4% and 99.5% on both primary folds, which suggests that many remaining errors come from the decision rule, such as a shifted class prior, rather than from how images are ranked.

The centralized model, trained once with its own training-set statistics, reaches 78.9% on CHENG and 51.9% on PMRAM, where it classifies a third of the tumors as notumor. Every recalibrated federated method exceeds it on both folds, and even with averaged statistics the federated means are comparable to it on CHENG and above it on PMRAM. With a single centralized seed we cannot say how much of this gap is seed variation, but centralized training is not an upper bound here: it too carries training-site statistics to the new site. The best single site is a strong baseline in one direction only: a model trained on CHENG alone reaches 84.1% on PMRAM, more than any recalibrated federated method and close to the best federated mean (84.9%), helped by never predicting notumor, a class it has not seen; one trained on PMRAM alone reaches 41.0% on CHENG. Internal validation is a poor guide to any of this: the centralized model of the SARTAJ fold scores 95.6% on held-out images of its training sites and 53.9% on SARTAJ.

C. Quantization cost

The interaction hypothesis predicts a larger FP32→INT8 loss after federated than after centralized training. With the statistics each model was trained with, the opposite holds on the only fold where the loss is large: the centralized model loses 8.6 points on PMRAM, whereas FedAvg loses -0.5 ± 2.5 and no federated method more than 4.6 ± 3.8 on average. On CHENG and SARTAJ no method loses more than 1.7 points on average, and FedBN with averaged statistics even gains 5.6 points on CHENG, a sign of how unstable those statistics are. Our data therefore do not support the hypothesis, with the caveat that the centralized estimate rests on one seed.

With recalibrated statistics QAF-Net has the smallest Δ_q on CHENG (0.15 ± 0.14 against 0.48–1.08) and on SARTAJ (0.12 ± 0.59 against 0.98–1.72), and a smaller Δ_q than FedProx, its $\lambda = 0$ ablation, on all three folds; on PMRAM FedAvg has the smallest (-0.55 ± 1.13 against 2.12 ± 1.36 for QAF-Net). Taken alone, this would support the regularizer, so we checked whether it acts as designed. If it pulled weights toward the INT8 grid, the residual $(w - Q(w))/s$, measured in quantization steps, would concentrate near zero. Fig. 2 shows it uniform for both methods over all nine trained models of each. Its mean square is 0.0818 for QAF-Net and 0.0817 for FedProx, which is what a uniform residual gives $(1/12)$ once the 1.8% of weights lying exactly on the grid, the per-channel maxima among them, are counted. The cause is scale: on the trained models $\lambda \mathcal{R}_q$ evaluates to about 1×10^{-5}, against an average task loss of about 0.23 in the final round, so its gradient is swamped. QAF-Net and FedProx were run with the same seeds and differ only in this term; the gap between them measures how sensitive federated training is to a small perturbation, not a property of the regularizer.

For FedProx and QAF-Net the quantization loss is largest on PMRAM. One candidate cause, which we have not tested, is that activation ranges are calibrated on training-site images while recalibrated statistics describe the target site.

D. Error structure

On PMRAM every federated method recognizes gliomas (97–99%) and most pituitary tumors but sends 43–53% of meningiomas to glioma (Fig. 3). The centralized model errs the other way: it keeps most meningiomas and pituitary tumors but calls 59% of gliomas notumor.

TABLE III

LEAVE-ONE-SITE-OUT RESULTS ON THE HELD-OUT COHORT, MEAN ± S.D. OVER THREE SEEDS. INT8 IS HELD-OUT ACCURACY (%) AFTER FULL-INTEGER CONVERSION; Δ_q IS THE FP32→INT8 ACCURACY LOSS IN POINTS (LOWER IS BETTER); MISSED IS THE PERCENTAGE OF TUMORS CLASSIFIED AS NOTUMOR; SUBTYPE IS INT8 ACCURACY ON TUMOR IMAGES USING ONLY THE THREE TUMOR LOGITS. CENTRALIZED AND LOCAL-ONLY ARE SINGLE RUNS; THE LOCAL-ONLY ROW IS THE BEST SINGLE TRAINING SITE FOR EACH FOLD, CHOSEN ON THE TEST COHORT (PMRAM, CHENG AND BR35H RESPECTIVELY).

	CHENG held out			PMRAM held out			SARTAJ held out		
Method	INT8	Δ_q	Missed	INT8	Δ_q	Missed	INT8	Δ_q	Subtype
No target data: batch-normalization statistics from the training sites (size-weighted client average when federated)									
Centralized	78.9	−0.98	9.0	51.9	8.62	33.8	53.3	0.55	68.9
Local-only, best site	41.0	5.06	0.0	84.1	2.68	0.0	63.2	−0.92	49.4
FedAvg [5]	76.7 ±17.7	−0.77 ±2.43	0.0 ±0.0	57.5 ±11.1	−0.53 ±2.47	0.0 ±0.0	24.1 ±2.6	1.65 ±1.12	55.7 ±5.9
FedProx [14]	74.0 ±29.2	−0.94 ±0.81	0.0 ±0.0	84.9 ±7.6	4.59 ±3.76	0.4 ±0.6	24.6 ±4.3	0.61 ±0.87	53.5 ±8.6
FedBN [15]	73.7 ±15.3	−5.59 ±4.25	0.0 ±0.0	76.5 ±20.0	−1.76 ±2.83	0.0 ±0.0	27.1 ±1.4	0.61 ±1.35	62.8 ±3.2
QAF-Net (ours)	77.5 ±1.8	−1.64 ±2.05	0.2 ±0.2	73.4 ±21.6	1.95 ±1.03	2.3 ±3.3	27.3 ±2.6	−0.12 ±0.38	62.4 ±7.3
Batch-normalization statistics recalibrated on unlabeled held-out scans									
FedAvg [5]	93.3 ±2.0	1.06 ±1.01	0.0 ±0.0	77.3 ±5.0	−0.55 ±1.13	0.9 ±0.3	25.0 ±2.0	1.72 ±2.02	57.9 ±4.6
FedProx [14]	94.2 ±1.0	0.48 ±0.37	0.7 ±0.4	64.9 ±13.1	4.17 ±1.46	12.6 ±12.6	30.0 ±1.3	1.59 ±1.01	65.4 ±3.7
FedBN [15]	94.4 ±0.7	1.08 ±0.23	0.0 ±0.0	78.3 ±2.9	0.86 ±2.57	0.3 ±0.2	28.2 ±0.6	0.98 ±0.70	65.2 ±1.3
QAF-Net (ours)	94.7 ±0.6	0.15 ±0.14	0.6 ±0.2	73.6 ±2.2	2.12 ±1.36	5.7 ±6.2	29.4 ±0.7	0.12 ±0.59	67.2 ±2.3

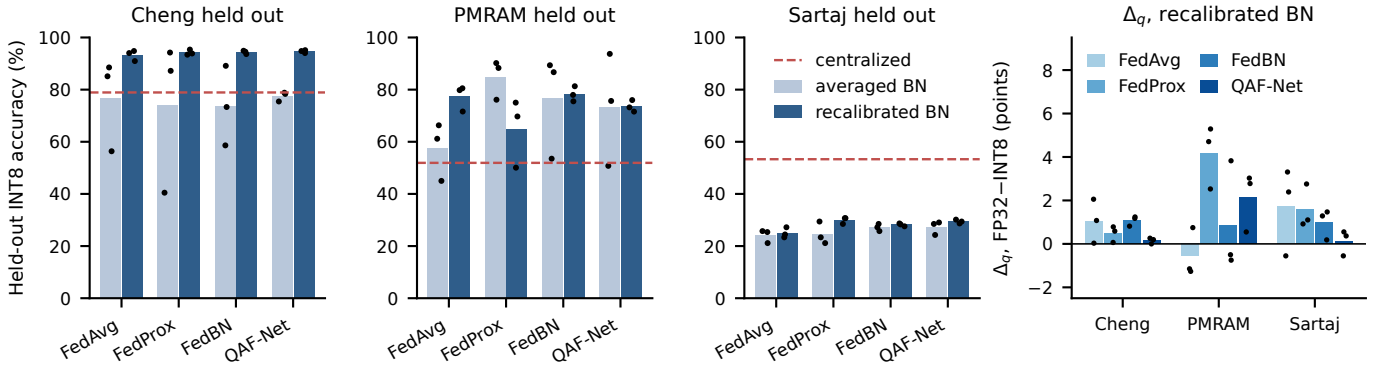


Fig. 1. Held-out INT8 accuracy by fold with the two sources of batch-normalization statistics (bars: mean over three seeds; dots: individual seeds; dashed line: centralized, one seed), and the FP32→INT8 loss Δ_q with recalibrated statistics.

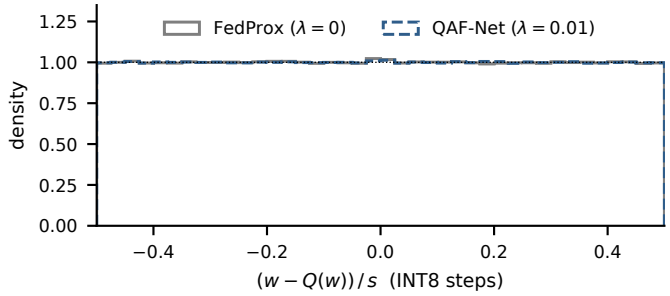


Fig. 2. Distance of each convolutional and dense weight from its INT8 grid point, in quantization steps, pooled over the nine trained models of each method. A working quantization regularizer would concentrate mass at zero; both distributions are uniform. Weights lying exactly on the grid (1.8%) are omitted.

On SARTAJ the federated methods reach 57.9–67.2% subtype accuracy, close to the centralized 68.9%, but only 25.0–30.0% overall, because they almost never predict notumor: over three seeds FedAvg and FedBN recognize none of the 309 normal scans, QAF-Net 0.6% and FedProx 3.0%. The centralized model recognizes 43.0%, and a model trained on BR35H alone reaches 63.2% overall accuracy. In this fold BR35H is the only client with normal scans and holds 19% of the training images; a class held by a single client is outweighed in the average, the failure that analyses of non-identically distributed federated training predict [26]. The sequence confound of Section III may explain why even the centralized model misses most normal scans, but it does not explain the gap between centralized and federated training.

E. Explainability

Fig. 4 shows Grad-CAM maps for QAF-Net on six PMRAM gliomas. The maps highlight the skull boundary and the background more often than the lesion, including on correctly classified slices. At 128×128 input the last convolutional map is 4×4, so localization is coarse, and six slices are a small sample; still, the maps give no evidence that the predictions rest on the lesion.

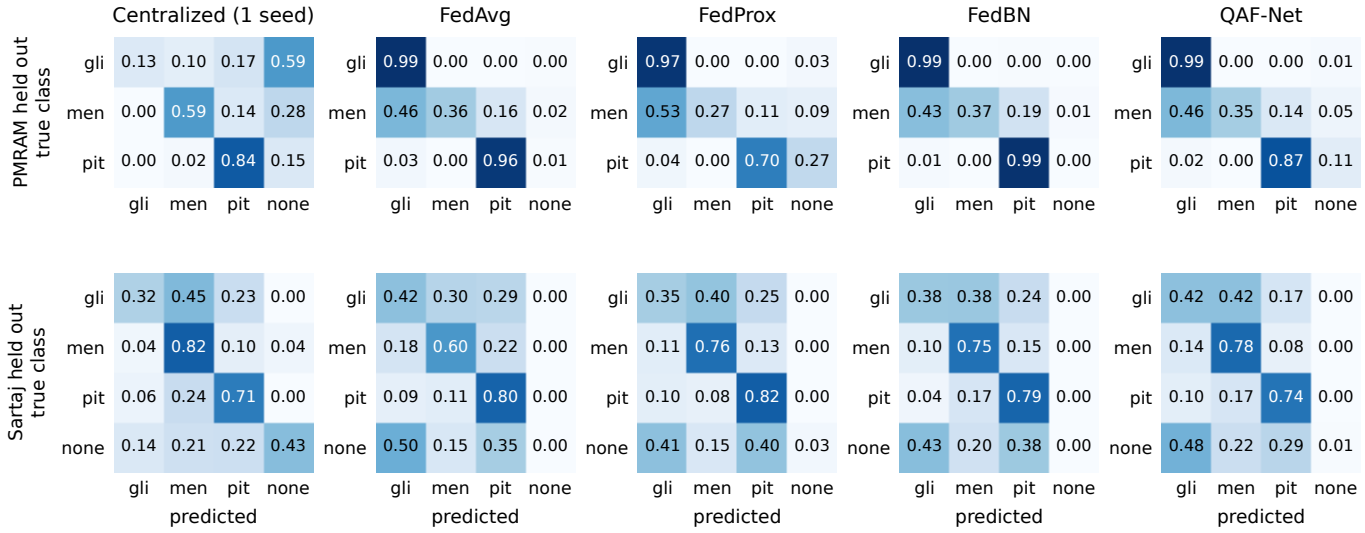


Fig. 3. Row-normalized INT8 confusion matrices on the PMRAM fold (top) and the SARTAJ fold (bottom). Federated methods use recalibrated statistics and are pooled over three seeds; the centralized model is a single run.

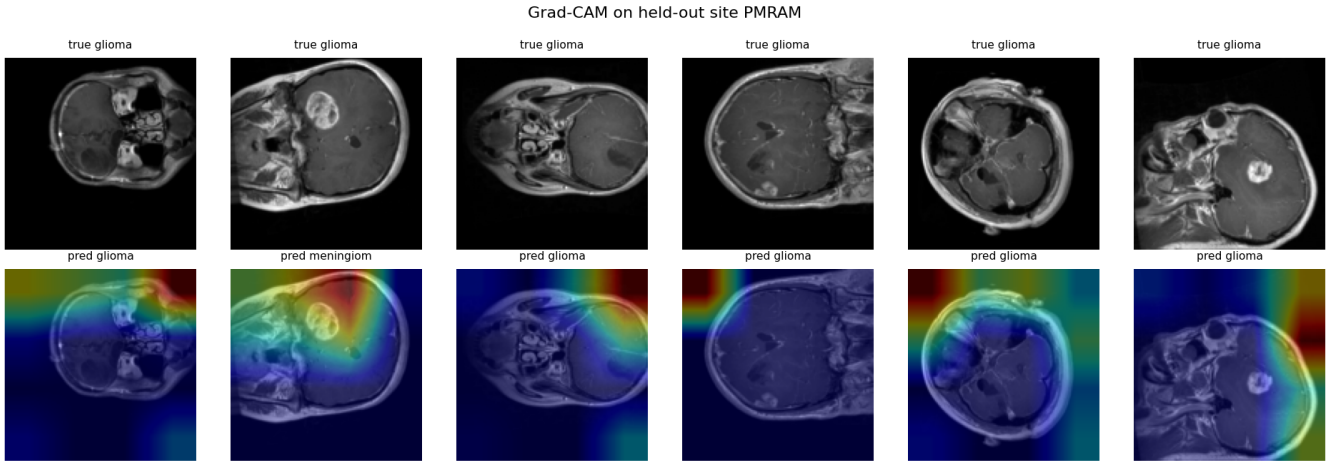


Fig. 4. Grad-CAM for QAF-Net (seed 0, averaged statistics) on six held-out PMRAM slices. Top: input and true class; bottom: attribution and predicted class.

TABLE IV
DEPLOYMENT PROFILE OF THE INT8 MODEL, WHICH IS THE SAME FOR EVERY TRAINING METHOD. LATENCY IS SINGLE-IMAGE INFERENCE WITH THE TENSORFLOW LITE INTERPRETER ON A KAGGLE CPU SESSION: MEAN OVER THE 36 FEDERATED ARTIFACTS, WITH THEIR RANGE IN PARENTHESES.

Parameters	415,332
Multiply–accumulates per image	18.9 M
Site-local batch-normalization scalars	28,160
INT8 `.tflite` size	606.3 KB
Tensor arena, estimated lower bound	$\geq$453.6 KB
CPU latency, mean	1.90 ms (1.86–1.96)
CPU latency, 95th percentile	2.34 ms (2.06–2.56)

F. Deployment

Every method produces the same artifact (Table IV), since the objectives and normalization choices change the weights and not the graph. The INT8 model is 606.3 KB and classifies a slice in about 1.9 ms on a CPU. It fits the 1 MB flash of an Arduino Nano 33 BLE, but its tensor arena, estimated from below at 453.6 KB, exceeds that board's 256 KB of SRAM and the 264 KB of a Raspberry Pi Pico. As built it targets microcontrollers with at least 512 KB of SRAM or application-class processors; the 256 KB class needs a smaller network or lower input resolution.

VII. CONCLUSION

We set out to test whether federated training makes brain tumor classifiers harder to quantize, and to counter the effect with a quantization-aware federated objective. Across three leave-one-site-out folds over four public cohorts, reduced by a duplicate audit to images that are genuinely distinct, the

effect did not appear: federated models lose about as much accuracy to full-integer quantization as a centrally trained one, or less. At the strength tested our regularizer does not move the weights toward the integer grid, so QAF-Net's small quantization loss on two folds cannot be credited to it.

Two findings do hold. First, how an unseen site obtains its batch-normalization statistics matters more than the federated algorithm: re-estimating them from unlabeled local scans lifts held-out accuracy by 17 to 21 points on CHENG and cuts the spread across seeds there from up to 29 points to at most 2, and the resulting 606 KB INT8 model classifies a slice in about 2 ms on a CPU. Second, federation loses a class that only one site holds: the best federated model recognized only 3% of the normal scans of the held-out cohort.

The next steps follow from these results: a regularizer scaled to the quantization step, or a quantized forward pass, with λ swept rather than fixed; recalibration on a sample separate from the evaluation images; class-aware aggregation for classes held by a single site; a smaller network for 256 KB-SRAM microcontrollers; and multi-seed centralized baselines with recalibration, so that the comparison with federation is like for like.

LIMITATIONS

The federation is simulated on a single machine; we do not measure communication cost, straggler behavior, or the effect of client dropout. All cohorts supply 2D slices rather than 3D volumes, and we evaluate a single architecture. Latency is measured on an x86 CPU as a proxy for microcontroller execution, and the reported tensor arena is a lower-bound estimate from the model's tensors rather than a measurement from an on-device run. Recalibration is transductive: it uses the unlabeled images it is then evaluated on. The centralized and local-only baselines are single runs without recalibration, three seeds are too few for significance tests, and λ was fixed rather than tuned. Patient-level grouping is available only for CHENG; for the remaining cohorts we cannot exclude the possibility that slices from one patient appear in both a training and a validation partition of the same site, though the leave-one-site-out results—our primary claim—are unaffected, since entire cohorts are held out. Public datasets carry label noise that our copy analysis only partly quantifies. Normal training scans come only from cohorts that include T2-weighted and FLAIR slices, so a model may partly associate acquisition sequence with the notumor class; subtype accuracy limits the consequence on the tumor-only folds but does not remove the confound. Finally, deduplication leaves only two public cohorts with substantial subtype labels, and four clients is a small federation; the heterogeneity effects we study would be better characterized with more sites.

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